# Sepsis & Shock – Chapter Summary

Definitions • Hemodynamics • Resuscitation • Vasopressors

ICU / EDIC Board Review • MCQ-Oriented

## 1. Sepsis Definitions (Sepsis-3)

- **Sepsis:** life-threatening organ dysfunction from dysregulated host response to infection
- **Organ dysfunction:** acute ↑ SOFA ≥ 2 (≈ 10% mortality)
- **Septic shock:** vasopressor to keep MAP ≥ 65 **AND** lactate > 2 mmol/L despite adequate fluids (mortality > 40%)
- **qSOFA** (RR ≥ 22, altered mentation, SBP ≤ 100): prognostic, NOT a screening tool alone (SSC 2021)
- SIRS no longer part of the definition

**Phoenix criteria (2024)** for pediatric sepsis

## 2. SOFA Score Components

| System | Variable |
|---|---|
| Respiratory | $PaO_2/FiO_2$ |
| Coagulation | Platelets |
| Liver | Bilirubin |
| Cardiovascular | MAP / vasopressors |
| CNS | GCS |
| Renal | Creatinine / urine output |

- Each scored 0–4; max 24
- Baseline SOFA assumed 0 unless known pre-existing dysfunction

## 3. Shock Classification

| Type | CO | SVR | Filling (CVP/PAWP) | $SvO_2$ |
|---|---|---|---|---|
| Hypovolemic | ↓ | ↑ | ↓ | ↓ |
| Cardiogenic | ↓ | ↑ | ↑ | ↓ |
| Obstructive | ↓ | ↑ | ↑ (RV) | ↓ |
| Distributive | ↑/N | ↓ | ↓/N | ↑/N |

- **Obstructive:** PE, tamponade, tension pneumothorax, auto-PEEP
- **Distributive:** sepsis, anaphylaxis, neurogenic, adrenal crisis, liver failure
- Mixed shock is common (septic cardiomyopathy)

## 4. Hour-1 Bundle (SSC)

- Measure **lactate**; remeasure if > 2
- **Blood cultures** before antibiotics (don't delay)
- **Broad-spectrum antibiotics**: within 1 h if shock or probable sepsis; within 3 h if possible sepsis without shock
- **30 mL/kg crystalloid** for hypotension or lactate ≥ 4 (weak recommendation)
- **Vasopressors** if MAP < 65 during / after fluids

**Source control** as soon as practical (ideally < 6–12 h)

## 5. Fluid Resuscitation

- **Balanced crystalloids** preferred over 0.9% saline (SMART, PLUS/BaSICS: neutral-to-modest benefit)
- **Albumin** if large volumes of crystalloid needed (ALBIOS)
- **Avoid:** starches (HES – AKI & mortality: 6S, CHEST), gelatins
- **Restrictive vs liberal** after initial bolus: no difference (CLOVERS, CLASSIC)

### Fluid responsiveness

- **PLR:** ↑ SV/CO ≥ 10% (works with arrhythmia & spontaneous breathing)
- **PPV / SVV > 12–13%:** only if controlled ventilation, VT ≥ 8, sinus rhythm, closed chest
- **End-expiratory occlusion test**, mini-fluid challenge, IVC variability (limited)
- **CRT** normalization as target (ANDROMEDA-SHOCK)

## 6. Vasopressors & Inotropes

| Drug | Receptor | Use |
|---|---|---|
| Norepinephrine | $\alpha_1 > \beta_1$ | First-line septic shock |
| Vasopressin | $V_1$ | Add at NE 0.25–0.5 µg/kg/min (VASST) |
| Epinephrine | $\beta_1, \beta_2, \alpha_1$ | 2nd add-on; ↑ lactate |
| Phenylephrine | $\alpha_1$ | Tachyarrhythmia |
| Dobutamine | $\beta_1$ | Low CO with adequate MAP |
| Angiotensin II | $AT_1$ | Refractory vasodilatory (ATHOS-3) |
| Methylene blue | NO/cGMP | Rescue |

- **MAP target 65** (higher MAP 80–85 no benefit – SEPSISPAM; 65-trial in ≥ 65 yrs: permissive OK)
- Peripheral norepinephrine is safe short term
- **Dopamine:** more arrhythmias (SOAP II) – avoid

## 7. Corticosteroids in Shock

- **Hydrocortisone 200 mg/day** if ongoing vasopressor requirement (NE ≥ 0.25 µg/kg/min ≥ 4 h)
- **ADRENAL:** faster shock reversal, no mortality ↓
- **APROCCHSS** (+ fludrocortisone 50 µg): ↓ 90-day mortality
- Don't use ACTH stimulation test to decide
- Side effects: hyperglycemia, hypernatremia

## 8. Antimicrobial Principles

- Empiric broad cover → de-escalate at 48–72 h
- **MRSA cover** if high risk; **double Gram-negative** only if high MDR risk
- **Prolonged infusion β-lactams** (BLING III: possible mortality benefit)
- **Procalcitonin** to guide discontinuation (not initiation)
- Shorter courses when source controlled (7 days)
- **Antifungal** only if high Candida risk (TPN, abdominal surgery, colonization)
- Pharmacokinetics: ↑Vd & augmented renal clearance → loading doses

## 9. Lactate & Perfusion

- **Type A:** hypoperfusion (shock, hypoxemia, anemia)
- **Type B:** liver failure, metformin, epinephrine / salbutamol, thiamine deficiency, malignancy, toxic alcohols, linezolid, propofol infusion syndrome
- Lactate-guided resuscitation: target ↓ 20% per 2 h
- **$ScvO_2$ ≥ 70%:** EGDT no benefit (ProCESS, ARISE, ProMISe)
- **Pcv-$aCO_2$ gap > 6 mmHg** → low flow state

## 10. Cardiogenic Shock

- **SCAI stages:** A (at risk), B (beginning), C (classic), D (deteriorating), E (extremis)
- **AMI-CS:** early revascularization (SHOCK trial); culprit-only PCI (CULPRIT-SHOCK)
- **Norepinephrine** first-line; dobutamine / milrinone for low CO
- **IABP:** no mortality benefit (IABP-SHOCK II)
- **Impella CP:** ↓ mortality in STEMI-CS (DanGer Shock) but ↑ complications
- **VA-ECMO:** no benefit routinely (ECLS-SHOCK)
- **Cardiac power output** < 0.6 W = poor prognosis
- **Pulmonary artery catheter**: CI < 2.2, PAWP > 15

## 11. Anaphylaxis

- **IM epinephrine 0.5 mg** (1:1000) anterolateral thigh; repeat q5 min
- Refractory → IV epinephrine infusion; glucagon if on β-blockers
- Large-volume fluids; airway early (angioedema)
- Antihistamines & steroids: adjunct only
- **Tryptase** within 1–2 h confirms
- Biphasic reaction: observe 6–24 h

## 12. Other Shocks

- **Neurogenic:** cord injury above T6 → hypotension + bradycardia; NE ± atropine
- **Adrenal crisis:** hydrocortisone 100 mg IV bolus then 200 mg/day
- **Toxic shock syndrome:** staph / strep; clindamycin + source control ± IVIG
- **Tamponade:** Beck's triad, pulsus paradoxus > 10, RA/RV diastolic collapse → pericardiocentesis
- **Thyroid storm / myxedema** as causes of shock

## 13. Septic Cardiomyopathy & Adjuncts

- Reversible biventricular dysfunction (7–10 days); ↓ EF with normal / low filling pressures
- **Levosimendan:** no benefit (LeoPARDS)
- **Esmolol / landiolol** for tachycardia: inconsistent; not routine
- **Vitamin C, thiamine (VITAMINS, LOVIT):** no benefit / harm
- **Blood:** transfuse Hb < 7 g/dL (TRISS)
- **Glucose** 144–180 mg/dL (NICE-SUGAR)
- **Bicarbonate:** pH ≤ 7.2 + AKI (BICAR-ICU)
- VTE + stress ulcer prophylaxis (if risk)

## 14. ECMO, Blood Purification & Others

- **Polymyxin B hemoperfusion:** no mortality benefit (EUPHRATES)
- **Cytokine adsorption (CytoSorb):** not recommended
- **IVIG:** not for adult sepsis (except TSS)
- **VA-ECMO in septic cardiomyopathy** with very low CI: possible in experienced centers

## 15. High-Yield MCQ Pearls

- [x] Septic shock = vasopressor for MAP ≥ 65 + lactate > 2
- [x] qSOFA is not a screening tool
- [x] Antibiotics ≤ 1 h in septic shock
- [x] Balanced crystalloid > 0.9% saline; avoid HES
- [x] PLR ↑ SV ≥ 10% = fluid responsive
- [x] PPV invalid with arrhythmia / spontaneous breathing / low VT
- [x] NE first-line; add vasopressin at 0.25–0.5
- [x] Hydrocortisone 200 mg/day if vasopressor needed
- [x] EGDT: no benefit
- [x] Lactate Type B: metformin, epinephrine, liver failure
- [x] Hb trigger 7 g/dL; glucose 144–180
- [x] IABP: no benefit in AMI shock

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